Lecture 26 – Lenz' Law

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2020-03-03

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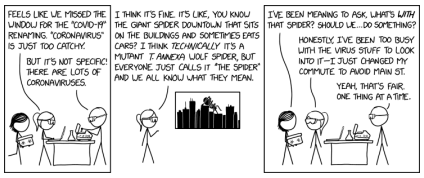

1 Faraday's Law Illustrations

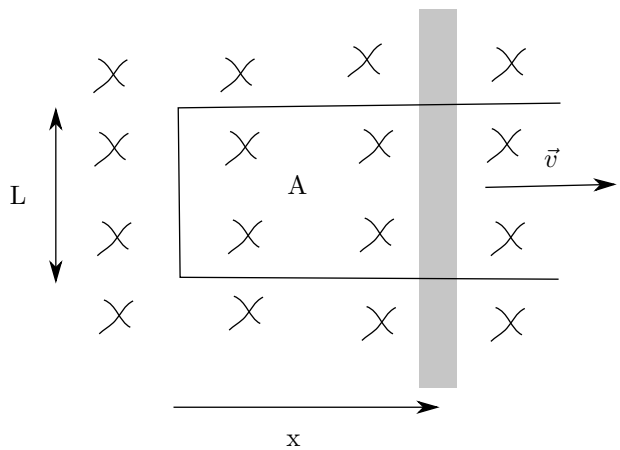

Figure 1: Current from changing loop size

Area is the area enclosed by the bar and the loop itself.

$$\Delta V = -\frac{d\phi}{dt}B$$

$$= -\frac{d}{dt}\vec{B}.\vec{A}$$

(A and B are parallel, resulting in following)

$$= -\frac{d}{dt}(BA)$$

B is constant in the following:

$$= -B\frac{dA}{dt}$$

$$A = Lx$$

$$\frac{dA}{dt} = L\frac{dx}{dt} = Lv$$

$$\Delta V = -BLv$$

This is also known as motional EMF.

2 Lenz' Law

Lenz' Law contends with the issue of determining direction of induced current. It's description is as follows:

"The current induced by Faraday's Law, I_{ind}, is such that it produces an induced magnetic field $\vec{B}_{ind}$ (equivalently ϕ_{ind}) that opposes the rate of change $\frac{d\phi_{ext}}{dt}$, of external flux, ϕ_{ext}."

Table 1: Direction of Current Above Example

Quantity	**Direction**
ϕ_{ext}	X
$\frac{d\phi_{ext}}{dt}$	X
ϕ_{ind}	O
I_{ind}	CCW

2.1 Another Faraday illustration

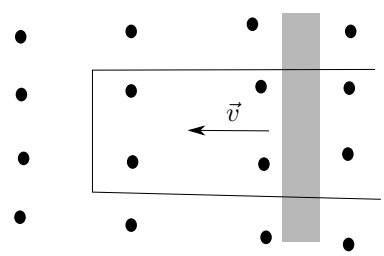

Figure 2: Inverted Faraday example

Table 2: Direction of Current Above Example

Quantity	**Direction**
ϕ_{ext}	O
$\frac{d\phi_{ext}}{dt}$	X
ϕ_{ind}	O
I_{ind}	CCW

2.2 The torus and the magnet

(Use your imagination to make this into an actual torus)

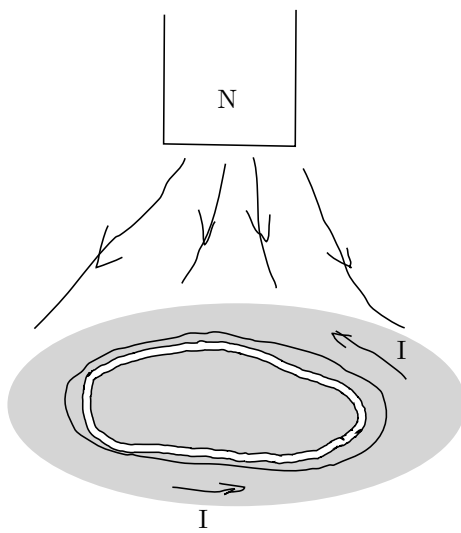

Figure 3: Torus and the magnet

Table 3: Direction of Current Above Example

Quantity	**Direction**
ϕ_{ext}	$\downarrow$
$\frac{d\phi_{ext}}{dt}$	$\downarrow$
ϕ_{ind}	$\uparrow$
I_{ind}	CCW